

Research Article

Multi-horizon prediction of rainfall-induced slope responses using physical state-space modeling

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ARTICLE INFO

Keywords:

Rainfall-induced slope displacement
Physical state space model
Extended Kalman filter
Multi-step forecasting
Uncertainty quantification

ABSTRACT

Accurate prediction of rainfall-induced slope response remains challenging for geological hazard early warning, especially in low-dimensional monitoring settings where conventional machine learning methods often struggle with multi-horizon forecasting and uncertainty quantification. This study proposes a prediction framework based on a two-state physical state-space model (SSM) combined with an Extended Kalman Filter (EKF). The rainfall-response process is represented by two physical latent states: soil saturation and proximity to a critical failure state. A delayed EKF scheme is introduced to maintain consistency between training and online inference, while analytical covariance propagation provides calibrated 90% prediction intervals. Using field tilt-monitoring data from a slope site in Japan, the proposed method is compared with a tree-based ensemble model (XGBoost) and a sequence model (LSTM). Across forecast horizons from 1 to 24 h, the SSM, with only 8 learnable parameters, consistently achieves higher R^2 than both baselines, reaching 0.857 for the downslope component at 12 h. Its 90% interval coverage remains stable at 91–93%, whereas XGBoost declines to 61% at 24 h. The learned wetness-decay parameter yields a data-driven half-life estimate consistent with empirical effective-rainfall formulations used in Japanese slope engineering, supporting the physical plausibility of the two-state decomposition. These results suggest that explicit physical latent-state modeling is effective for multi-horizon slope-response prediction under rainfall-driven monitoring conditions.

1. Introduction

Rainfall-induced slope instability causes thousands of fatalities and substantial economic losses worldwide each year (Froude & Petley, 2018). Because slope deformation is a progressive process, displacement typically begins to accumulate hours before macroscopic failure occurs, and high-frequency continuous observations from tilt sensors provide the data foundation needed to detect these precursory signals and enable early warning (Stähli et al., 2015; Uchimura et al., 2015). Nevertheless, how to effectively integrate sensor readings, historical rainfall records, and meteorological forecast information to accurately predict slope displacement over multi-step horizons spanning several hours or longer, while reliably quantifying predictive uncertainty, remains a core unresolved challenge.

Two main research directions have been pursued to address this problem. Physics-based numerical simulations explicitly model rainfall infiltration and pore-pressure evolution, offering good interpretability, but require extensive site investigation data for parameter calibration and incur prohibitive computational costs that preclude real-time online

warning applications (Park et al., 2013; Raia et al., 2014). Empirical approaches such as rainfall threshold methods are computationally lightweight but cannot capture site-specific hydrological responses (Segoni et al., 2018). Machine learning approaches instead learn predictive rules directly from historical rainfall and sensor time series; representative strategies include sequence models that operate on sliding time windows (Li et al., 2021; Nava et al., 2023) and tree-based ensemble methods driven by manually engineered effective rainfall indices (Ma et al., 2022; Nakane et al., 2023). These approaches have demonstrated encouraging results for short-horizon displacement prediction, yet existing work is almost entirely confined to single-step-ahead forecasting. Genuine multi-step horizon modeling remains largely absent, and this limitation is not coincidental: it is rooted in structural characteristics intrinsic to the landslide prediction problem.

The limitations of machine learning in this specific prediction setting do not stem from algorithmic deficiencies per se, but rather from three structural features of the problem that cannot be overcome by scaling model capacity or stacking additional features. First, raw sensor readings contain significant high-frequency noise, and pure regression frameworks cannot distinguish physical signal from noise during

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Received 12 April 2026; Received in revised form 2 July 2026; Accepted 2 July 2026

Available online 5 July 2026

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Nomenclature

Symbol	Description
$\mathbf{x}_t$	Latent state vector $[h_{\text{wet},t}, h_{\text{inst},t}]^\top$
h_{wet}	Wetness state: rainfall accumulation-and-decay proxy
h_{inst}	Instability state: proximity to response activation
$a_{\text{wet}}, a_{\text{inst}}$	Decay coefficients of h_{wet} and h_{inst}
c_{sc}	Coupling coefficient $h_{\text{wet}} \rightarrow h_{\text{inst}}$
c_{fut}	Forecast rainfall influence on h_{inst}
k_{obs}	Observation scaling coefficient
$\sigma_{\text{wet}}, \sigma_{\text{inst}}, \sigma_{\text{obs}}$	Process and observation noise standard deviations
$r_t, r_{\text{fut},t}$	Current and mean future rainfall
$y_t, \hat{y}_t$	Observed and predicted tilt-angle increment
H	Prediction horizon (hours)
$\mathbf{F}, \mathbf{J}^h$	State transition and observation Jacobians
$\mathbf{P}_{t s}, \mathbf{K}_t$	State covariance and Kalman gain
Δ	Delay length in the Delayed EKF

optimization, meaning the learning target itself is contaminated (Zhang et al., 2017). Second, the rainfall-displacement response exhibits threshold nonlinearity: when pore pressure remains below a critical level, the slope shows virtually no systematic deformation, whereas displacement accelerates sharply once that level is exceeded. Gradient-based optimization applied to the severely imbalanced sample distribution, which consists of many quiescent periods and few active events, tends to average out rare surge events and cannot reliably capture this activation behavior (He & Garcia, 2009; Lin et al., 2017). Third, in such monitoring settings the observable signals are limited to two low-dimensional channels, namely rainfall and sensor tilt angle, and the statistical learning efficiency of overparameterized models is fundamentally mismatched to this information budget (Nakkiran et al., 2020; Vapnik, 1999). Handcrafted feature engineering based on multi-half-life effective rainfall sequences merely conceals this fundamental deficit rather than resolving it. The common root cause underlying all three limitations is that the true physical state driving the displacement response, specifically the degree of soil moisture saturation and accumulated proximity to a critical failure state, is never explicitly modeled and remains hidden inside the black box (Karpatne et al., 2017; Reichstein et al., 2019).

State-space modeling and Kalman filtering are established tools for dynamic state estimation (Durbin, Koopman, 2012; Thrun et al., 2005), and monitoring-based early-warning frameworks are already used in landslide and geohazard studies (Intrieri et al., 2012; Stähli et al., 2015). The contribution of this study is therefore more specific: a compact two-state physical latent representation, a Delayed EKF for information-flow consistency in multi-step prediction, and analytical covariance propagation for uncertainty quantification. To address the limitations above, this paper proposes a multi-step slope displacement prediction framework based on a physics-based state-space model (SSM) combined with an extended Kalman filter (EKF). The model explicitly represents soil moisture saturation and critical state proximity as two latent states, characterizes the complete rainfall-to-displacement transmission chain with a small number of physically interpretable parameters, and incorporates future rainfall information as a known external input in the state transition, thereby enabling forward-looking multi-step prediction. For inference, a Delayed EKF scheme is proposed to resolve the information misalignment between training and online prediction in Multi-horizon forecasting, ensuring strict consistency in information flow and providing prediction intervals through analytical propagation of state covariance. The framework is evaluated on real tilt sensor data from a slope monitoring site in Japan, using gradient-boosted tree methods and sequence models as baselines, with

systematic assessment of prediction accuracy and uncertainty coverage across multiple forecast horizons.

The rest of this paper is organized as follows. Section 2 introduces the methodology, including the two-state SSM structure, the Delayed EKF inference framework, and the parameter optimization strategy. Section 3 describes the study site and dataset. Section 4 presents the quantitative results. Section 5 discusses model interpretability and predictive performance. Section 6 concludes the paper.

2. Methodology

2.1. Problem formulation and methodological framework

Under sustained rainfall, rainfall-prone slopes may progressively approach a critical state, eventually triggering progressive displacement (Iverson, 2005; Leroueil, 2001). Tilt sensors record slope surface angle at fixed time intervals; the increment between successive readings directly reflects the deformation rate and therefore constitutes the primary signal for early warning (Uchimura et al., 2015). Accordingly, accurately predicting the sensor's angular increment over a future horizon is equivalent to making a short-term quantitative estimate of the slope deformation trend (Uchimura et al., 2015).

The objective of this study is to develop a predictive model that uses historical rainfall observations and sensor readings up to the current time, together with a rainfall forecast over a future horizon, to estimate the angular increment of the tilt sensor over the next H hours:

$$\hat{d}_{\text{sensor}}(t, H) = f_{\theta}(\{R_{\tau}\}_{\tau \leq t}, \{\text{sensor}(\tau)\}_{\tau \leq t}, \{R_{\tau}\}_{\tau \in (t, t+H]}) \quad (1)$$

where R_{τ} denotes rainfall intensity at time τ , $\text{sensor}(\tau)$ denotes the tilt sensor reading, and the prediction horizon satisfies $H \in \{1, 2, \dots, 24\}$ (hours). θ is the learnable parameter set of the model (Table 1). The prediction target is defined as:

$$d_{\text{sensor}}(t, H) = \text{sensor}(t+H) - \text{sensor}(t) \quad (2)$$

This study predicts the tilt-angle increment $d_{\text{sensor}}(t, H)$ of a surface tilt sensor under rainfall forcing. This task is referred to as tilt response prediction. It differs from displacement prediction, which estimates spatial movement using GNSS/InSAR (Crosta, Agliardi, 2003; Intrieri et al., 2012), and from slope failure prediction, which detects catastrophic-onset transitions (Fukuzono, 1985; Saito, 1961).

To achieve this objective, a two-state physics-based state space model (SSM) is proposed. This model decomposes the process linking rainfall to slope angular change into two latent states with explicit physical interpretations, which are connected to sensor observations through a nonlinear observation function consistent with soil mechanics failure mechanisms. State estimation and prediction are performed using an Extended Kalman Filter (EKF). The overall system architecture is shown in Fig. 1; the design of each component is detailed in the following sections.

Table 1

Learnable parameters and system hyperparameters of the two-state SSM.

Parameter	Type	Physical interpretation
a_{wet}	Learnable	Decay coefficient of h_{wet} (soil drainage rate)
c_{sc}	Learnable	Coupling coefficient $h_{\text{wet}} \rightarrow h_{\text{inst}}$
c_{fut}	Learnable	Influence weight of forecast rainfall on h_{inst}
a_{inst}	Learnable	Decay coefficient of h_{inst} (critical state dissipation rate)
k_{obs}	Learnable	Observation scaling coefficient
σ_{wet}	Learnable	Process noise standard deviation of h_{wet}
σ_{inst}	Learnable	Process noise standard deviation of h_{inst}
σ_{obs}	Learnable	Observation noise standard deviation
$\theta_0 = 2$	hyperparameter	Activation threshold of the observation function
$\alpha = 1.33$	hyperparameter	Power exponent of the observation function (degree of nonlinearity; determined by cross-validation on the training set)

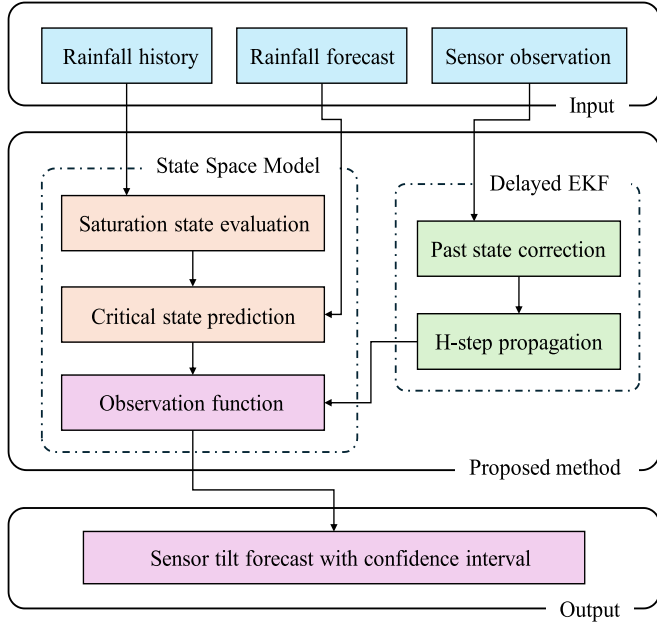


Fig. 1. System architecture of the proposed SSM-EKF prediction framework.

2.2. Two-State physics-based SSM

The central design principle is that each component of the model, the latent states, the state transition equations, and the observation equation, corresponds directly to the physical processes governing rainfall-induced slope tilt response prediction, rather than serving as a purely data-driven fitting structure.

The latent state vector is defined as:

$$\mathbf{x}_t = \begin{bmatrix} h_{wet,t} \\ h_{inst,t} \end{bmatrix} \quad (3)$$

where h_{wet} represents the wetness state encoding rainfall accumulation and decay, and h_{inst} represents the instability state tracking proximity to a response-triggering threshold. Known external inputs include the current-step rainfall r_t and the mean forecast rainfall over the next H steps, $r_{fut,t} = \text{mean}(r_{t+1}, \dots, r_{t+H})$.

The two latent states h_{wet} and h_{inst} are interpretable variables designed to approximate the relevant physical processes governing rainfall-induced tilt response; they do not directly correspond to measured hydrological variables or strict soil-mechanical state variables. Specifically, h_{wet} encodes a rainfall accumulation-and-decay process analogous to effective rainfall and soil wetting (Eq. (4)), while h_{inst} aggregates wetting history and forecast rainfall into a scalar state that tracks proximity to a response-triggering threshold (Eq. (5)). The threshold observation function then links h_{inst} to accelerated tilt response (Eq. (6)). Both states are learned from rainfall and tilt observations through the EKF framework (Sections 2.3–2.4) and should be interpreted as functional proxies rather than direct estimates of soil moisture, pore-water pressure, groundwater level, factor of safety, or shear strength.

The transition equation for the saturation state h_{wet} is:

$$h_{wet,t} = a_{wet} \cdot h_{wet,t-1} + r_t + w_{wet,t} \quad (4)$$

where $a_{wet} \in (0, 1)$ is a decay coefficient and $w_{wet,t} \sim \mathcal{N}(0, \sigma_{wet}^2)$ is process noise. h_{wet} increases with rainfall input r_t and decays exponentially at rate a_{wet} in the absence of rainfall, representing the natural drainage of the soil (Iverson, 2005). Mathematically, this state is equivalent to effective rainfall defined under a certain half-life, except that the half-life is not preset but determined from the data (Kosugi

et al., 2013).

The transition equation for the critical state h_{inst} is:

$$h_{inst,t} = c_{sc} \cdot h_{wet,t} + c_{fut} \cdot r_{fut,t} + a_{inst} \cdot h_{inst,t-1} + w_{inst,t} \quad (5)$$

where c_{sc} is the transfer coefficient from the wetness state to the instability state, c_{fut} quantifies the influence of future rainfall, a_{inst} is the dissipation rate of the instability state, and $w_{inst,t} \sim \mathcal{N}(0, \sigma_{inst}^2)$ is process noise. The state h_{inst} is driven jointly by current saturation and the rainfall forecast, and dissipates at rate a_{inst} . Incorporating future rainfall as an input is a natural modeling choice, because the prediction target is itself the sensor reading H steps ahead.

The observation function maps h_{inst} to the sensor measurement:

$$y_t = k_{obs} \cdot \text{softplus}(h_{inst,t} - \theta_0)^\alpha + v_t \quad (6)$$

where $\text{softplus}(x) = \ln(1 + e^x)$ (Glorot et al., 2011), k_{obs} is a learnable scaling coefficient, θ_0 is an activation threshold, α is the power exponent, and $v_t \sim \mathcal{N}(0, \sigma_{obs}^2)$ is observation noise. The design of this function reflects the threshold behavior characteristic of slope failure: when h_{inst} remains below θ_0 , the slope is in a stable state and the sensor exhibits negligible systematic change; once h_{inst} exceeds θ_0 , the slope enters the near-critical regime and the tilt response intensifies rapidly with increasing h_{inst} . This behavior is consistent with the physical picture of a slope that undergoes rapid deformation only after sufficient energy has accumulated (Iverson, 2005; Leroueil, 2001) (Fig. 2).

The model has 8 learnable parameters and 2 system hyperparameters, summarized in Table 1.

θ_0 and α are both fixed as hyperparameters rather than optimized jointly with the other parameters. Sensitivity analysis over $\theta_0 \in \{0, 0.5, 1, 2, 3, 4\}$ with five random seeds per value shows that $\{\theta_0, k_{obs}, c_{fut}, c_{sc}\}$ behave as a parameter equivalence class: changes in θ_0 are largely absorbed by compensatory changes in the learned parameters, whereas $\theta_0 = 0$ can lead to degenerate convergence with inflated k_{obs} and a_{inst} at its lower bound. Validation runs with learnable α across $H = 3, 6, 8, 12$, and 24 h with five seeds per horizon show that successful runs usually converge near 1.33, but making α learnable increases the probability of local-optimum or boundary solutions at the shortest and longest horizons. Based on these results, this study adopts $\theta_0 = 2$ and $\alpha = 1.33$.

2.3. EKF inference and the delayed update scheme

The Extended Kalman Filter (EKF) is the standard online inference framework for nonlinear state space models (Thrun et al., 2005). It alternates between a prediction step and an update step to produce

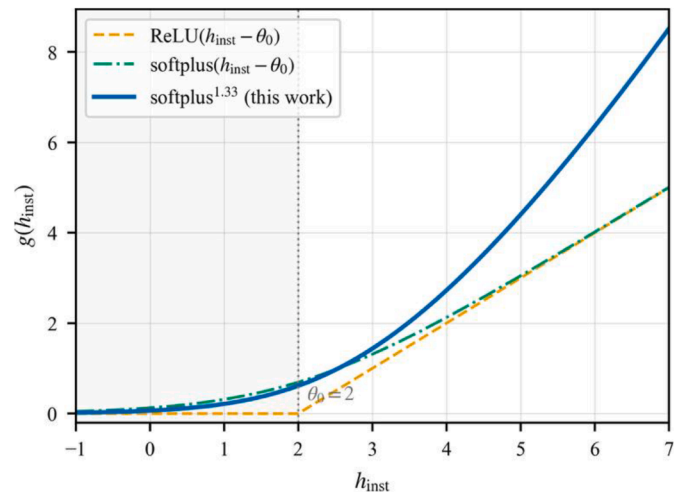


Fig. 2. Observation function: power-softplus compared with softplus and ReLU, with activation threshold.

real-time estimates of the latent state. In the notation below, $\hat{\mathbf{x}}_{t|s}$ denotes the estimate of the state at time t given information up to time s , and $\mathbf{P}_{t|s}$ is the corresponding covariance matrix.

The standard EKF executes the following at each time step t :

$$\hat{\mathbf{x}}_{t|t-1} = \mathbf{f}(\hat{\mathbf{x}}_{t-1|t-1}, \mathbf{r}_t, \mathbf{r}_{\text{fut},t}), \quad \mathbf{P}_{t|t-1} = \mathbf{F}_t \mathbf{P}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q} \quad (7)$$

$$\mathbf{K}_t = \mathbf{P}_{t|t-1} (\mathbf{J}_t^h)^\top (\mathbf{J}_t^h \mathbf{P}_{t|t-1} (\mathbf{J}_t^h)^\top + \sigma_{\text{obs}}^2)^{-1} \quad (8)$$

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t (\mathbf{y}_t - g(\hat{\mathbf{x}}_{t|t-1})), \quad \mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t \mathbf{J}_t^h) \mathbf{P}_{t|t-1} \quad (9)$$

where $\mathbf{f}(\cdot)$ is the state transition function, $\mathbf{Q} = \text{diag}(\sigma_{\text{wet}}^2, \sigma_{\text{inst}}^2)$ is the process noise covariance, $\mathbf{K}_t$ is the Kalman gain, $g(\cdot)$ is the observation function (Eq. (6)), and $\mathbf{F}_t$ and $\mathbf{J}_t^h$ are the Jacobians of $\mathbf{f}$ and g with respect to $\mathbf{x}$, respectively. Because the state transition equations are linear in $\mathbf{x}$, the Jacobian $\mathbf{F}$ (which is time-invariant) can be computed analytically:

$$\mathbf{F} = \begin{bmatrix} a_{\text{wet}} & 0 \\ c_{\text{sc}} \cdot a_{\text{wet}} & a_{\text{inst}} \end{bmatrix} \quad (10)$$

The off-diagonal term $c_{\text{sc}} \cdot a_{\text{wet}}$ captures the chained influence of the saturation state on the accumulation of the critical state. Because g depends on $\mathbf{x}$ only through h_{inst} , the Jacobian $\mathbf{J}^h$ takes the form:

$$\mathbf{J}^h = \begin{bmatrix} 0 & \frac{\partial g}{\partial h_{\text{inst}}} \end{bmatrix}^\top \quad (11)$$

The prediction target $\mathbf{y}_t = d_{\text{sensor}}(t, H)$ depends on sensor readings over the interval $[t, t + H]$. However, in an online setting, this target can only be used to update the state retrospectively after the full interval has elapsed. If the standard EKF update step (Eqs. (8)–(9)) is applied directly during training, training effectively uses future information that is unavailable at inference time. This constitutes an information misalignment (i.e., a structural discrepancy in the information available during training versus online inference), inflating apparent training performance (Fig. 3).

To resolve this issue, the update step is delayed by Δ steps, where Δ is the number of time steps corresponding to the prediction horizon H . The prediction step (Eq. (7)) proceeds normally for all t . The update step is applied only at $\tau = t - \Delta$, at which point the sensor window $[\tau, \tau + H]$ that defines $\mathbf{y}_\tau = d_{\text{sensor}}(\tau, H)$ has only just become available at the current time t (since $\tau + H = t$). The update equations are structurally identical to the standard case (Eqs. (8)–(9)), with subscripts replaced by τ :

$$\mathbf{K}_\tau = \mathbf{P}_{\tau|\tau-1} (\mathbf{J}_\tau^h)^\top (\mathbf{J}_\tau^h \mathbf{P}_{\tau|\tau-1} (\mathbf{J}_\tau^h)^\top + \sigma_{\text{obs}}^2)^{-1} \quad (12)$$

$$\hat{\mathbf{x}}_{\tau|\tau} = \hat{\mathbf{x}}_{\tau|\tau-1} + \mathbf{K}_\tau (\mathbf{y}_\tau - g(\hat{\mathbf{x}}_{\tau|\tau-1})), \quad \mathbf{P}_{\tau|\tau} = (\mathbf{I} - \mathbf{K}_\tau \mathbf{J}_\tau^h) \mathbf{P}_{\tau|\tau-1} \quad (13)$$

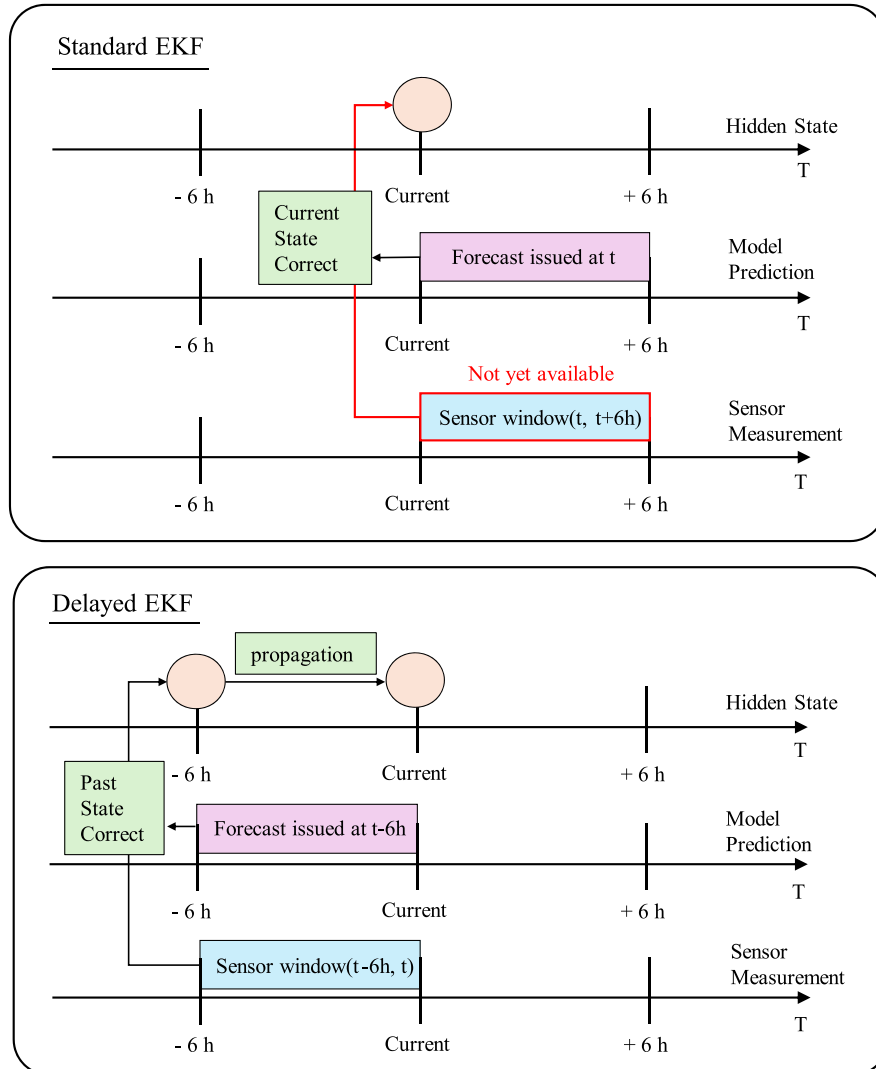


Fig. 3. Standard EKF versus Delayed EKF inference schemes.

With this design, the information available during training and inference is strictly consistent, eliminating the misalignment by construction. This scheme is structurally equivalent to the sample-state augmentation approach described in the delayed-measurement EKF literature (Gopalakrishnan et al., 2011); under the fixed-delay condition of the present study, it reduces to the simplified form of directly correcting the lagged state and forward-propagating to the current estimate.

After the update at τ , the state covariance $\mathbf{P}_{t|\tau}$ is propagated forward through H prediction steps (Eq. (7)) to obtain $\mathbf{P}_{t|t+H}$, which accumulates H steps of process noise. This differs from the one-step-ahead covariance in the standard EKF: the longer the horizon H , the greater the uncertainty accumulated in $\mathbf{P}_{t|t+H}$. Linearizing through the observation function yields the forecast variance in the observation space:

$$\sigma_{y,t}^2 = \mathbf{J}_t^H \mathbf{P}_{t|t+H} (\mathbf{J}_t^H)^T + \sigma_{\text{obs}}^2 \quad (14)$$

Under a Gaussian assumption, the 90% prediction interval is:

$$\text{PI}_{90\%} = [\hat{y}_t - 1.645 \sigma_{y,t}, \hat{y}_t + 1.645 \sigma_{y,t}] \quad (15)$$

where $\hat{y}_t = g(\hat{\mathbf{x}}_{t|\tau})$ is the forecast mean and $\sigma_{y,t}$ is the forecast standard deviation (Eq. (14)).

2.4. Model learning

Within the EKF framework, parameter learning is equivalent to maximizing the log-likelihood over the training set (Durbin & Koopman, 2012). Under the Gaussian assumption, the negative log-likelihood (NLL) decomposes into the one-step prediction error at each time step:

$$\mathcal{L}(\theta) = \sum_{t=1}^T \left[\frac{(y_t - \hat{y}_t)^2}{2\sigma_{y,t}^2} + \frac{1}{2} \ln \sigma_{y,t}^2 \right] \quad (16)$$

where $\theta = \{a_{\text{wet}}, c_{\text{sc}}, c_{\text{fut}}, a_{\text{inst}}, k_{\text{obs}}, \sigma_{\text{wet}}, \sigma_{\text{inst}}, \sigma_{\text{obs}}\}$ denotes all learnable parameters. The NLL acts as an automatic regularizer on the forecast variance: any attempt to inflate $\sigma_{y,t}^2$ to reduce the squared error term is penalized by the log-variance term, compelling the model to find an optimal trade-off between accuracy and reliability.

A warm-up period is applied before the NLL is accumulated: the EKF runs forward for a fixed number of initial steps to allow the latent states to converge from their zero initialization to physically plausible values, and these steps are excluded from the NLL summation. Without this warm-up, the bias from the initial state would dominate the early loss and distort parameter convergence. A warm-up length of 7 days is recommended.

Parameter optimization is performed using the L-BFGS-B algorithm (Byrd et al., 1995) as implemented in SciPy. This quasi-Newton method is well suited to the moderate number of parameters (8). Parameter constraints are enforced through differentiable reparameterization transforms (Table 2). Decay coefficients are mapped to their admissible intervals by scaled sigmoid transforms, while k_{obs} and the noise standard deviations are kept positive by exponential transforms. The core EKF

Table 2

Parameter constraints and reparameterization transforms used during optimization.

Parameter	Admissible range	Reparameterization
a_{wet}	[0.8, 0.9999]	scaled sigmoid
a_{inst}	[0.8, 0.9999]	scaled sigmoid
c_{sc}	unconstrained	identity
c_{fut}	unconstrained	identity
k_{obs}	>0	exponential
σ_{wet}	>0	exponential
σ_{inst}	>0	exponential
σ_{obs}	>0	exponential

loop is accelerated via Numba JIT compilation (Lam et al., 2015).

No normalization or standardization is applied to rainfall inputs or tilt-sensor readings before model training. Tilt increments are multiplied by a fixed constant 10^4 inside the implementation to improve numerical conditioning of k_{obs} and the observation-noise term; all reported RMSE and MAE values are converted back to degrees. This fixed unit scaling is not data-dependent normalization.

All experiments were conducted on an Intel Core i9-12900 K. Under the training configuration used in this study, optimization for a single prediction horizon takes approximately 5 min; the 24 horizons for each sensor axis can be optimized in parallel. For the representative case of the 7-hour horizon, both training and test NLL decrease rapidly within approximately 10 iterations, converge within 20–30 iterations, and the training and test curves remain closely aligned throughout, indicating no evidence of overfitting (Fig. 4).

3. Study site and dataset

The monitoring site investigated in this study is located near the prefectural boundary between Tokushima and Kochi Prefectures, Japan. The instrumented slope is situated on a valley side within mountainous terrain. It extends approximately 80 m along the slope direction, and about 40 m across the slope. The slope gradient within the monitored area is approximately 32° . Prior field and remote-sensing observations at this site have suggested that the slope exhibits persistent landslide activity. Comparison of aerial images taken over successive years shows surface-scale collapse features on the moving mass, as well as morphological changes that are consistent with ongoing downslope deformation. Kinematically, the landslide body can be divided along the slope-parallel direction into three zones: an upper tension zone, a middle translation zone, and a lower compression zone. In the translation zone, slope-parallel movement is the dominant kinematic signature. These three zones together reflect the internal displacement field of a coherent mass undergoing progressive creep. The tilt sensors analysed in the present study are installed within the translation zone. This installation position is consistent with the dominance of the downslope component dX reported in Section 3. The site has therefore been subject to prolonged, rainfall-driven slow creep, and is a suitable site for continuous displacement prediction research (Fig. 5). The continuous monitoring data used in this study span approximately 18 months, from June 2023 to December 2024.

Tilt data were collected by tilt sensors installed within the slope (Hafidz et al., 2023; Putra et al., 2021) at a sampling interval of 10 min and an angular resolution of 0.001° . Readings were recorded along both the X-axis (downslope) and the Y-axis (cross-slope). Rainfall data were obtained from the Japan Meteorological Agency AMeDAS network and composited from the three nearest observation stations using

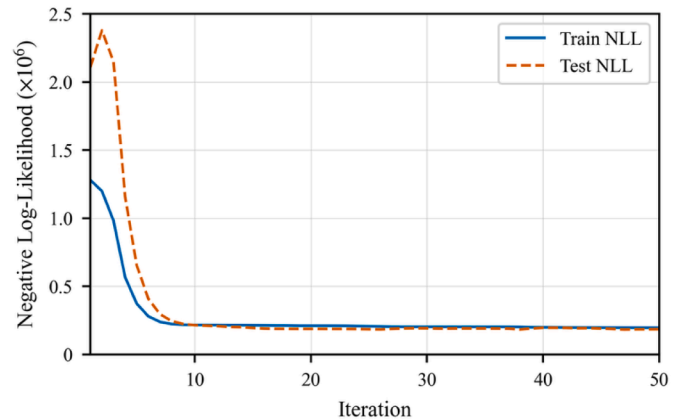


Fig. 4. Training learning curve: NLL versus L-BFGS-B iteration.

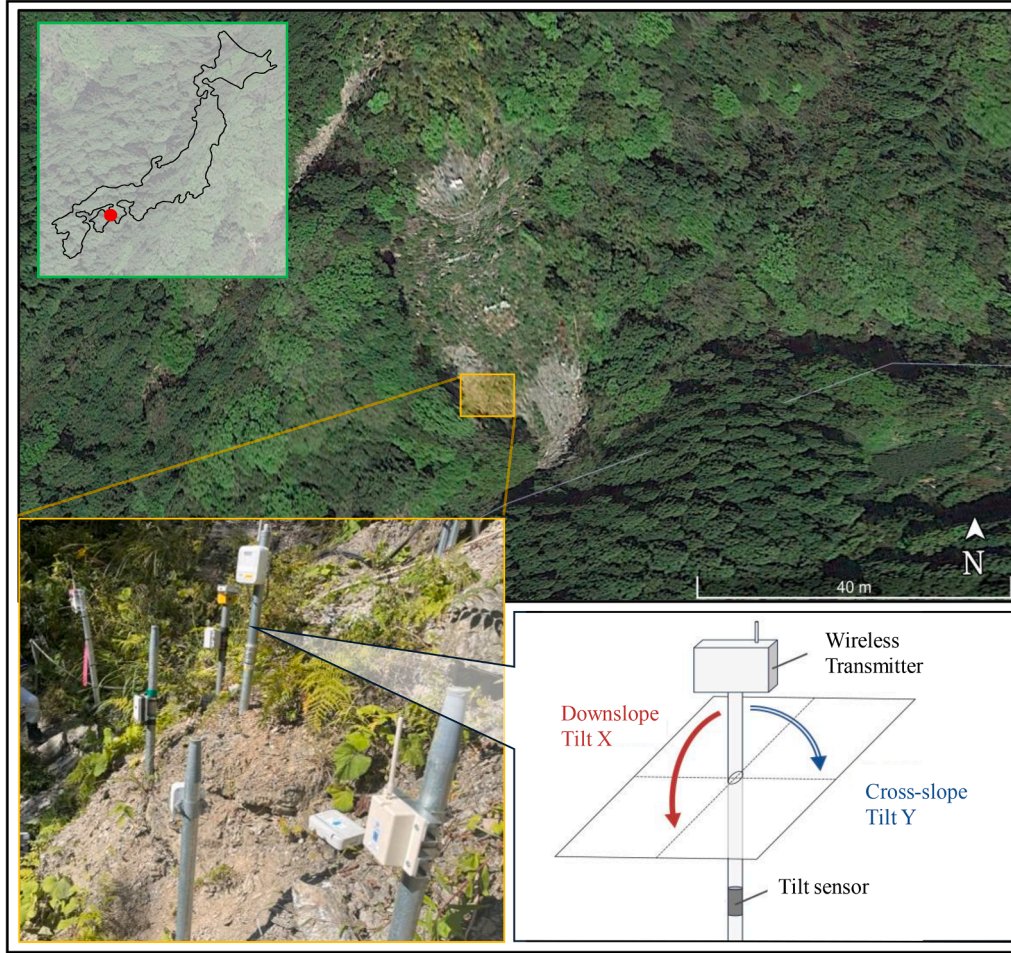


Fig. 5. Overview of the study site: location, aerial view (Google Earth (Google LLC, 2024)), field installation, and sensor axes).

inverse-distance weighting, then interpolated to the same 10-min resolution:

$$R(t) = \frac{\sum_{i=1}^3 w_i R_i(t)}{\sum_{i=1}^3 w_i}, \quad w_i = \frac{1}{d_i} \quad (17)$$

where $R_i(t)$ is the rainfall at station i at time t , and d_i is the distance from that station to the site. The future mean rainfall $r_{\text{fut},t}$ defined in §2.2 is computed from these observed AMeDAS records rather than from operational weather forecasts; consequently, the prediction performance reported in this study represents an upper bound under ideal rainfall-forecast conditions. The sensitivity of the model to rainfall input errors is examined in §5.3.

The prediction target is the tilt increment defined in Eq. (2), specified separately for the X and Y directions as:

$$dX_H(t) = X(t+H) - X(t), \quad dY_H(t) = Y(t+H) - Y(t) \quad (18)$$

This quantity represents the cumulative angular tilt response of the sensor relative to its current position over the future horizon of H hours. Each prediction horizon $H \in \{1, 2, \dots, 24\}$ corresponds to an independent prediction target series; dX and dY are modeled separately in this study.

The model inputs consist solely of two raw time series: tilt angle and rainfall. The tilt sensors recorded two-axis data at a sampling interval of 10 min and an angular resolution of 0.001° ; rainfall records from three AMeDAS stations were composited by inverse-distance weighting (Eq. (17)) and interpolated to the same 10-min temporal resolution. No missing-value imputation, outlier removal, filtering, smoothing, or

sensor-drift correction was applied before training; records within the data gap described below were excluded rather than interpolated. This raw-data setting was used to match the intended deployment condition; noise and residual uncertainty are handled within the model through the observation-noise term and learnable covariance parameters (§2.4). The resulting increment series dX and dY are approximately independently and identically distributed in a statistical sense, which makes them suitable for machine learning training and evaluation. Accordingly, the dataset was partitioned chronologically into a training set and a test set; all methods learned their parameters on the training set and were evaluated independently on the test set (Table 3):

A data gap of approximately 19 days separates the two periods; records within this gap are excluded from both sets. Across the 24 prediction horizons ($H \in \{1, \dots, 24\}$), a total of 48 independent prediction target series are constructed for each dataset (24 horizons for dX and 24 for dY). A full-period overview of the data is shown in Fig. 6.

Fig. 6 reveals several key characteristics of the dataset. Both the absolute tilt angles X and Y exhibit monotonically increasing trends, reflecting the continuous slow creep of the slope. The total range of

Table 3

Summary of the training and test dataset partition.

Dataset	Time period	Number of samples	Resolution
Training set	2023-06-05 to 2024-03-13	40,525	10-min interval, 0.001° angular
Test set	2024-04-01 to 2024-12-10	36,511	10-min interval, 0.001° angular

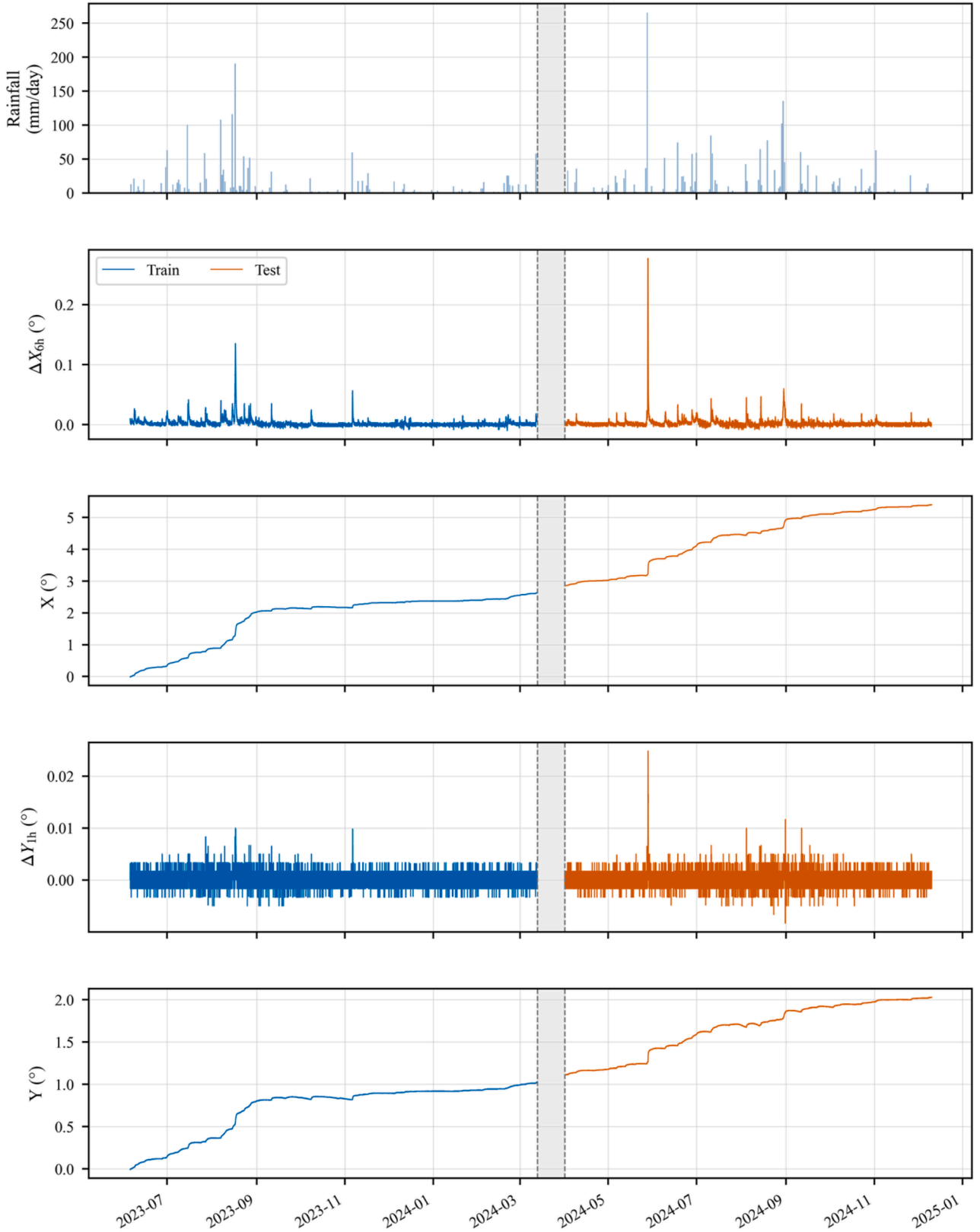


Fig. 6. Overview of the dataset: rainfall, tilt increments (dX , dY), and absolute tilt angles over the full monitoring period. Blue: training set; orange: test set.

variation in the X direction (downslope) substantially exceeds that in the Y direction (cross-slope), consistent with the primary sliding direction of the slope. In terms of increment series, dX and dY are broadly positively correlated with rainfall, and heavy rainfall events are generally

accompanied by larger displacement increments. However, for rainfall events of comparable intensity, the sensor response varies considerably across different periods: some events trigger pronounced displacement, while others produce little to no response. This complex, history-

dependent relationship between rainfall and tilt response is the central challenge in rainfall-driven tilt response prediction, and it provides the fundamental motivation for introducing latent state variables representing soil saturation and proximity to a critical failure state in the proposed framework. Furthermore, at the 1-hour prediction horizon, the dY series shows clearly visible high-frequency noise and negative oscillations, indicating that the cross-slope direction has a low signal-to-noise ratio at short timescales.

4. Results

4.1. Baseline methods

To objectively evaluate the proposed SSM, two categories of baseline methods were selected for systematic comparison: an ensemble tree model based on hand-crafted features and a sequential deep learning model. Both baselines were configured to produce uncertainty estimates, enabling a fair comparison across three metrics: R^2 , RMSE, and 90% prediction interval (CI) coverage rate.

XGBoost (Chen & Guestrin, 2016; Friedman, 2001) takes 73 hand-engineered features as input: 72 effective rainfall features corresponding to exponentially weighted cumulative rainfall with half-lives $M = 1$ h to 72 h, plus the mean forecast rainfall over the future H -hour horizon. The effective rainfall is computed via the following recurrence:

$$X_t^{(M)} = X_{t-1}^{(M)} \cdot e^{-\Delta t / M} + R_t \cdot e^{-\Delta t / M}, \quad \beta = \frac{\ln 0.5}{M} \quad (19)$$

where $\Delta t = 1/6$ h (10-minute time steps), M is the half-life in hours, and R_t is the rainfall at the current time step. This feature engineering requires 72 half-life values to be specified a priori based on domain knowledge, and remains fixed after training. Model hyperparameters were set to $n_{\text{estimators}} = 500$, $\text{max_depth} = 6$, and learning rate 0.05. The 90% prediction interval was obtained via quantile regression (Chen et al., 2022; Koenker & Jr., 1978): two separate XGBoost models were trained independently for $\alpha \in \{0.05, 0.95\}$, each minimising the pinball loss:

$$\mathcal{L}_\alpha(y, \hat{q}) = \sum_t \rho_\alpha(y_t - \hat{q}_\alpha(\mathbf{x}_t)), \quad \rho_\alpha(u) = u(\alpha - 1_{[u < 0]}) \quad (20)$$

The 90% CI is then $[\hat{q}_{0.05}(\mathbf{x}), \hat{q}_{0.95}(\mathbf{x})]$. Because the three models are trained independently, there is no guarantee that $\hat{q}_{0.05} \leq \hat{y} \leq \hat{q}_{0.95}$ (quantile crossing), which can cause the point estimate to fall outside the reported CI during extreme rainfall events.

LSTM (Hochreiter & Schmidhuber, 1997) uses a 72-hour sliding window (432 time steps at 10-minute resolution) as input, with four input dimensions: instantaneous rainfall, historical dX increments, historical dY increments, and the mean forecast rainfall over the future horizon. The 90% prediction interval is estimated from the residual standard deviation on the training set as $\pm 1.645 \times \sigma_{\text{residual}}$.

The input structure and uncertainty estimation approach for all three methods are summarised in Table 4:

All methods were evaluated on the same train/test split and all received identical future rainfall information as input, thereby eliminating information asymmetry as a confounding factor. Subsequent sections discuss each aspect in turn. Both baselines were trained on the full training set and evaluated on the held-out test set. No separate validation split, grid search, or early stopping was used; the reported configurations were selected in preliminary configuration checks and

then fixed for the comparison. For XGBoost, the settings were 500 trees, maximum depth 6, learning rate 0.05, row sub-sampling ratio 0.8, column sub-sampling ratio 0.8, and minimum child weight 5. The two quantile-regression models for the 90% prediction interval ($\alpha = 0.05$ and $\alpha = 0.95$) used the same tree configuration and were trained independently. For LSTM, the network comprised two stacked LSTM layers with hidden dimension 64 and inter-layer dropout 0.2, followed by one linear output unit (50,753 trainable parameters). Training used Adam with initial learning rate 10^{-3} , batch size 512, gradient clipping at maximum norm 1.0, ReduceLROnPlateau scheduling (patience 5, factor 0.5), and 50 epochs. All LSTM input features were standardised by per-channel z-score normalisation computed from training-set statistics.

4.2. Prediction accuracy

Fig. 7 and Fig. 8 illustrate how prediction accuracy varies with forecast horizon (1–24 h) for all three methods in the dX and dY directions.

SSM achieves systematically higher R^2 than both XGBoost and LSTM across the full 1–24 h range. Table 5 reports the corresponding RMSE, MAE, and R^2 values for both dX and dY at representative horizons. In the dX direction, SSM R^2 increases from 0.447 at 1 h to 0.857 at 12 h and remains at 0.822 at 24 h; its RMSE is also lower than both baselines at all listed horizons. XGBoost R^2 reaches only 0.338, 0.519, 0.563, and 0.642 at 1, 6, 12, and 24 h, respectively. LSTM partially closes the R^2 gap at 12 and 24 h, but its values remain below SSM's 0.857 and 0.822.

The SSM advantage is most pronounced at the 3–6 h range, where its R^2 is approximately 1.6 times that of XGBoost. This reflects the structural benefit of the two-state physical formulation over fixed half-life hand-crafted features in capturing the intermediate-timescale rainfall accumulation process.

Overall accuracy is lower in the dY direction than in dX, though SSM retains its leading position. The dX direction corresponds to the primary slope-parallel tilt response component (approximately 73% of total tilt response magnitude), while dY is smaller in magnitude and exhibits a lower signal-to-noise ratio at short horizons; this accuracy gap between directions is therefore physically expected. SSM dY R^2 is only 0.166 at 1 h but rises to 0.749–0.783 at horizons of 6 h and beyond, where the cumulative rainfall effect becomes the dominant driver and predictability improves substantially.

A consistent pattern is apparent in Fig. 8 across all three methods: R^2 rises steeply at short horizons (1–5 h) and then largely plateaus, with SSM stabilising quickly and the baselines still improving slowly thereafter but with limited additional gain. This behaviour has a clear physical basis: the process by which rainfall induces soil saturation, drives the slope toward a critical state, and ultimately produces measurable tilt response is not instantaneous (Iverson, 2005; Leroueil, 2001). Rather, a lag exists between the onset of rainfall and systematic deformation. At short horizons (1–3 h), all methods are constrained by a low signal-to-noise ratio, making it difficult to extract reliable signals from background noise. As the horizon extends beyond 5 h, the accumulated rainfall effect becomes dominant and model performance stabilises. Balancing prediction accuracy against warning lead time, a forecast horizon of 5–8 h appears to be the most practically effective range for SSM on this site, as R^2 is already close to its maximum within this window while still providing sufficient warning time.

These results also demonstrate that SSM achieves the best accuracy across all horizons with only 8 learnable parameters. This directly

Table 4

Comparison of input structure, model complexity, and uncertainty estimation approach for the three prediction methods.

Method	Input	Learnable parameters	CI estimation
SSM (proposed)	Rainfall + sensor observations + future H -hour mean	8	Analytical EKF covariance propagation
XGBoost	Effective rainfall ($M = 1$ –72 h) + future H -hour mean	3×500 trees (depth ≤ 6)	Quantile regression (independently trained)
LSTM	72-h sliding window rainfall + future H -hour mean	50, 753	Fixed residual $\pm 1.645\sigma$

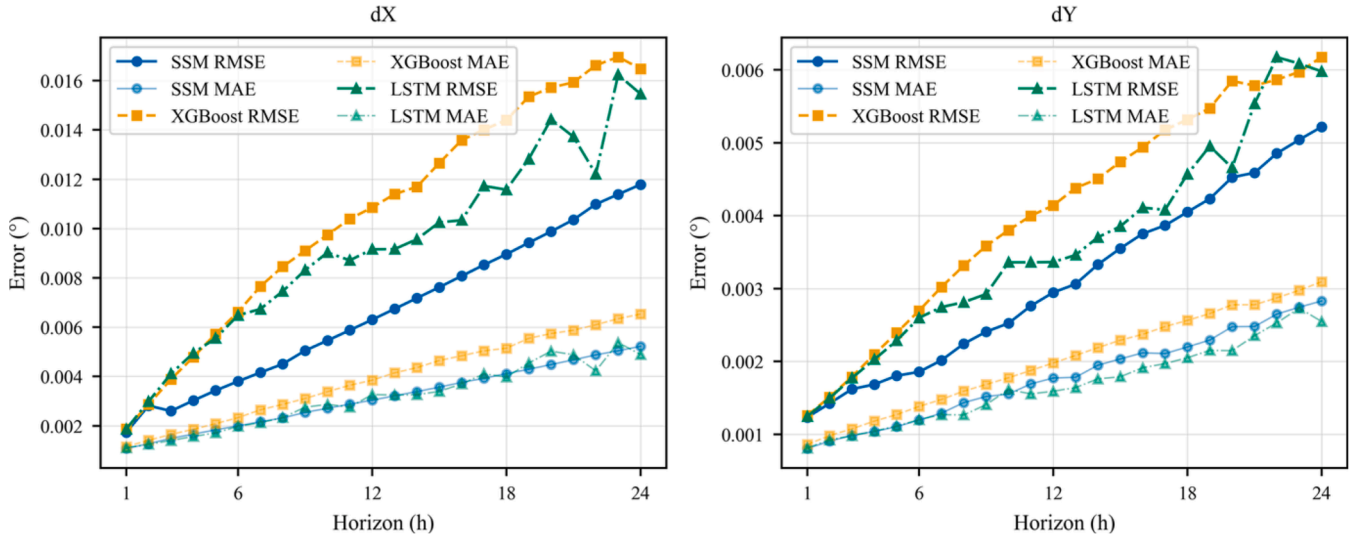


Fig. 7. Test-set RMSE and MAE versus prediction horizon (1–24 h) for SSM, XGBoost, and LSTM: dX (left) and dY (right).

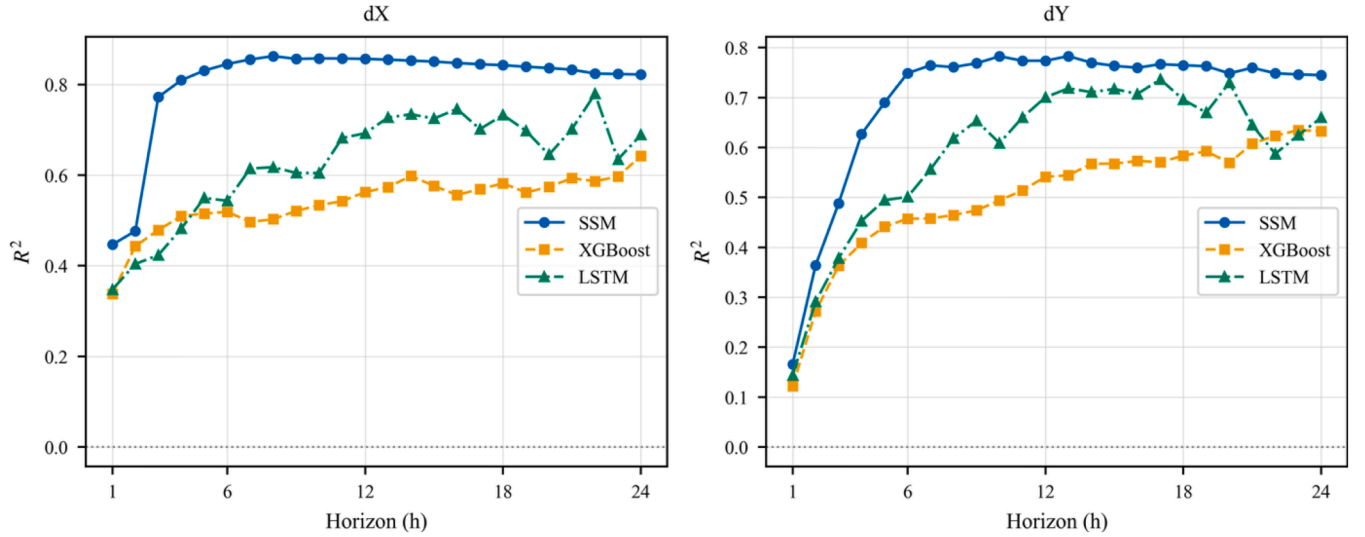


Fig. 8. Test-set R^2 versus prediction horizon (1–24 h) for SSM, XGBoost, and LSTM: dX (left) and dY (right).

Table 5

Test-set comparison (dX direction) across selected forecast horizons.

Horizon	Metric	SSM	LSTM	XGBoost
1 h	RMSE (deg)	0.00173	0.00187	0.00188
	MAE (deg)	0.00108	0.00111	0.00115
	R^2	0.447	0.347	0.338
3 h	RMSE (deg)	0.00260	0.00411	0.00389
	MAE (deg)	0.00149	0.00140	0.00165
	R^2	0.772	0.423	0.479
6 h	RMSE (deg)	0.00380	0.00648	0.00662
	MAE (deg)	0.00200	0.00198	0.00233
	R^2	0.845	0.543	0.519
12 h	RMSE (deg)	0.00631	0.00916	0.01086
	MAE (deg)	0.00304	0.00327	0.00385
	R^2	0.857	0.692	0.563
24 h	RMSE (deg)	0.01179	0.01546	0.01649
	MAE (deg)	0.00523	0.00490	0.00653
	R^2	0.822	0.689	0.642

corroborates the structural limitation identified in the introduction: given the low-dimensional monitoring signals of rainfall and sensor tilt angles, over-parameterised models cannot be adequately constrained by

the available data. Explicit physical state representation, rather than increasing model capacity, is the fundamental route to improved prediction performance.

4.3. Uncertainty estimation (90% prediction intervals)

Whether the 90% CI coverage rate is close to its nominal level is the primary criterion for evaluating the quality of uncertainty estimates (Fig. 9, Fig. 10).

SSM coverage rates remain close to the nominal level across all horizons (dX: 91–93%; dY: 88.9–95.1%), and interval widths expand adaptively with horizon length through EKF analytical propagation of the state covariance P . XGBoost severely underestimates uncertainty at long horizons: coverage falls monotonically from 83.3% at 1 h to 61.0% at 24 h, meaning that nearly 40% of true values fall outside the reported 90% CI. LSTM derives its CI from the per-horizon residual standard deviation ($\pm 1.645 \times \sigma_{\text{residual}}$); although width grows slightly with horizon, the growth rate is far smaller than that of SSM and does not respond to real-time rainfall intensity, resulting in a 24-h dX coverage of 81.6%. Overall, SSM provides the most reliable uncertainty estimates of the three methods through analytical covariance propagation.

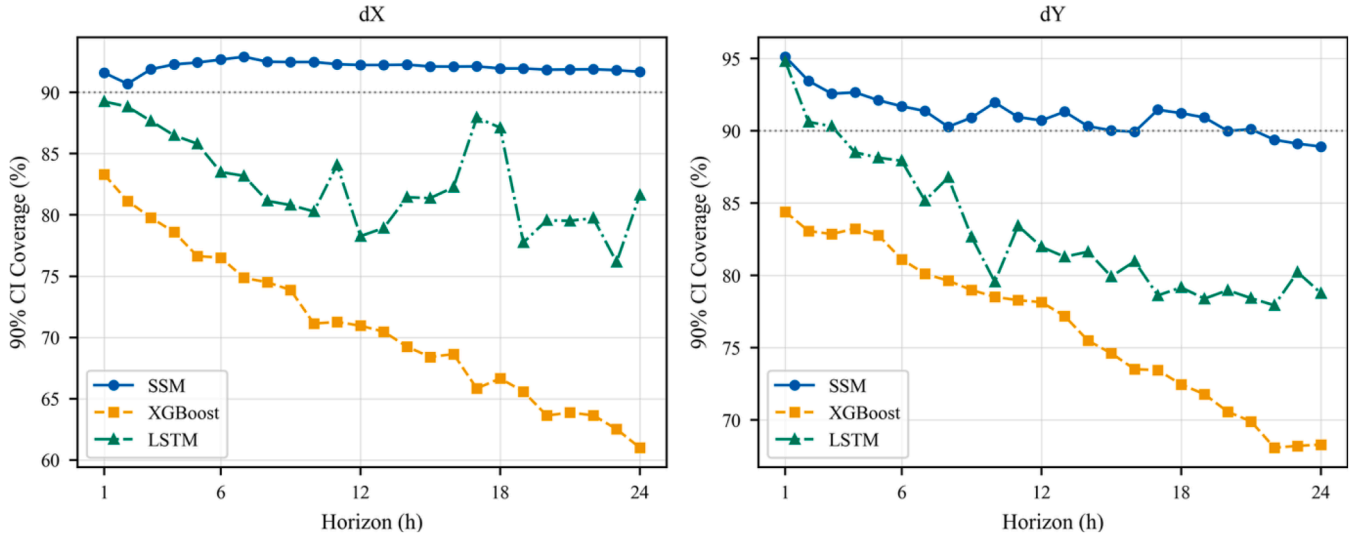


Fig. 9. Test-set 90% CI coverage rate versus prediction horizon (1–24 h): dX (left) and dY (right).

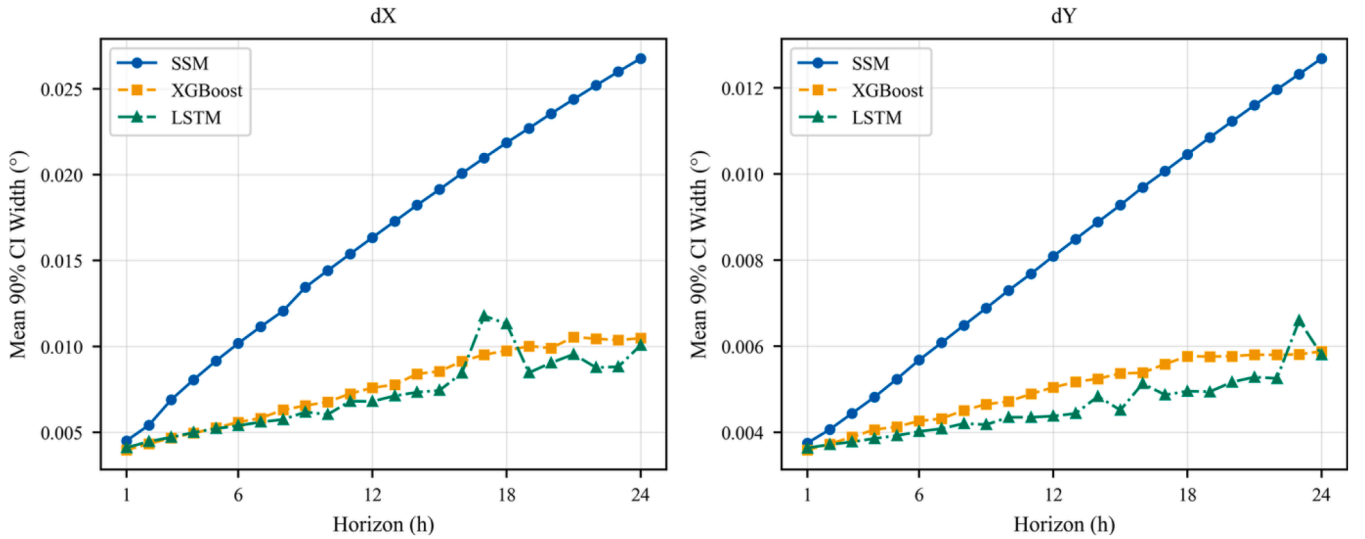


Fig. 10. Mean 90% CI width versus prediction horizon (1–24 h): dX (left) and dY (right).

4.4. Event-Level prediction comparison

Rainfall events were defined from daily cumulative rainfall: a day exceeding 30 mm was marked as a trigger day, trigger days separated by two or fewer days were merged, and each event window was extended to 25 h after the last trigger day to capture post-rainfall response. This identified 17 events covering approximately 18% of the test period.

For the pooled event-period data, MAE was decomposed into over-prediction and under-prediction components: $O = \frac{1}{N} \sum \max(\hat{y}_i - y_i, 0)$ and $U = \frac{1}{N} \sum \max(y_i - \hat{y}_i, 0)$, so that $MAE = O + U$. The ratio U/MAE quantifies prediction bias: values above 0.5 indicate under-prediction-dominated error, which is less favourable for early warning.

Peak-response accuracy was evaluated per event. For event k , the peaks y_k^{peak} and $\hat{y}_k^{\text{peak}}$ are the observed and predicted values at the time step of maximum absolute response. The peak R^2 across all 17 events is:

$$R_{\text{peak}}^2 = 1 - \frac{\sum_k (\hat{y}_k^{\text{peak}} - \bar{\hat{y}}_k^{\text{peak}})^2}{\sum_k (y_k^{\text{peak}} - \bar{y}_k^{\text{peak}})^2} \quad (21)$$

Table 6 summarises the pooled event-period results for dX. At $H = 6$ h, SSM achieves event-period $R^2 = 0.854$ and peak $R^2 = 0.904$, compared with 0.504/0.511 for LSTM and 0.481/0.578 for XGBoost. The under-prediction fraction reveals a clear bias difference: SSM has

Table 6

Pooled event-period metrics for dX across three forecast horizons ($H = 3, 6, 8$ h). MAE in $\times 10^{-3}$ deg. $U/MAE > 50\%$ indicates under-prediction-dominated error. Bold denotes best result per row.

	H	SSM	LSTM	XGBoost
Pooled R^2	3	0.814	0.387	0.480
	6	0.854	0.504	0.481
	8	0.865	0.584	0.449
Peak R^2	3	0.863	0.248	0.462
	6	0.904	0.511	0.578
	8	0.927	0.617	0.638
MAE	3	2.83	2.42	3.01
	6	4.64	4.23	5.24
	8	5.77	5.19	7.04
U/MAE	3	26.6%	56.5%	32.1%
	6	22.9%	54.7%	29.8%
	8	21.1%	53.3%	27.7%

$U/MAE = 22.9\%$, whereas LSTM has 54.7% . Most SSM error during events comes from over-prediction (conservative), while LSTM error is dominated by under-prediction of tilt response magnitude.

Two representative one-week windows at a 6-hour forecast horizon are selected to compare the predictive behaviour of each model under contrasting rainfall regimes. Both figures share the same layout: rainfall time series at the top, followed by the predicted versus observed tilt response (with 90% CI) for SSM (proposed), LSTM, and XGBoost in successive panels.

Case I (Fig. 11): Large slope tilt response triggered by a single intense rainfall event. During the early Meiyu season from 2024-05-25 to 2024-06-01, intense rainfall was concentrated within approximately three hours on 05-28. Observed dY tilt response rose rapidly to a peak of approximately 0.09° on 05-28/05-29 and then decayed gradually. SSM accurately tracks both the peak magnitude and timing as well as the subsequent decay, with the 90% CI closely bracketing the observed curve throughout. LSTM produces virtually no meaningful response during the event, with a predicted peak of only approximately 0.02° and

an extremely narrow CI that entirely misses the observed values. XGBoost similarly underestimates the event severely, with a peak below 20% of the observed value and a CI that fails to provide coverage. In this case, the SSM physical hidden state approaches the critical state rapidly under the intense short-duration rainfall, directly triggering the tilt response prediction and suggesting a clear advantage in handling sudden, high-intensity events.

Case II (Fig. 12): Continuous deformation under sustained multi-pulse rainfall. During the typhoon season from 2024-08-26 to 2024-09-02, rainfall arrived in repeated pulses over an extended period. Observed dX tilt response formed a double-peaked structure around 08-30 to 08-31, with peak values of approximately 0.06° . SSM tracks the overall trend reasonably well, with adaptively expanding CIs, although the inter-peak trough is slightly smoothed, the initial rise is marginally delayed, and the tail after 09-01 is mildly overestimated; these are inherent limitations of the two-state simplified model under complex multi-pulse forcing. LSTM broadly captures the deformation trend but with pronounced oscillations in the predicted curve and an

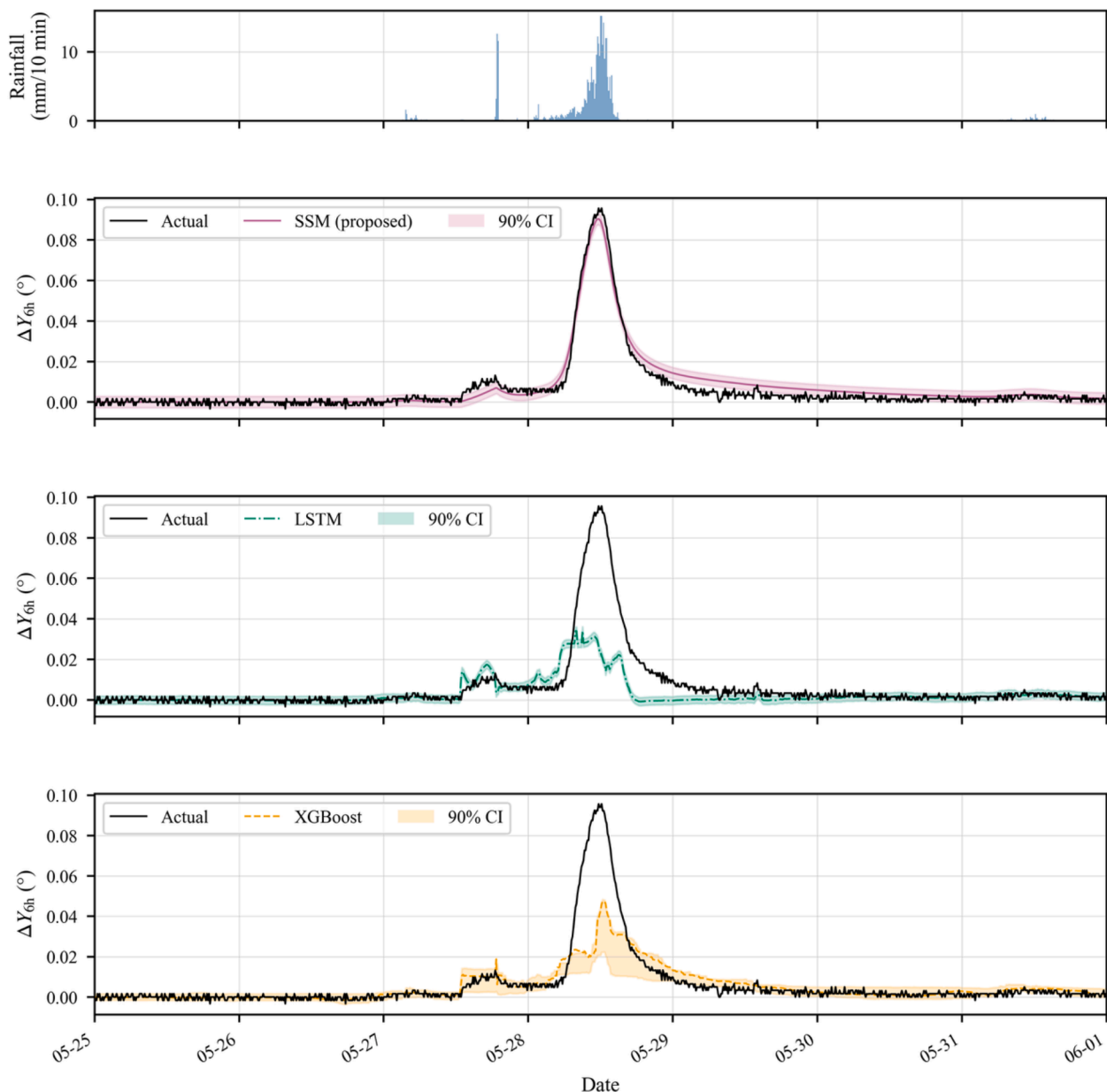


Fig. 11. Six-hour-ahead dY prediction comparison during a concentrated rainfall event (2024-05-25 to 2024-06-01).

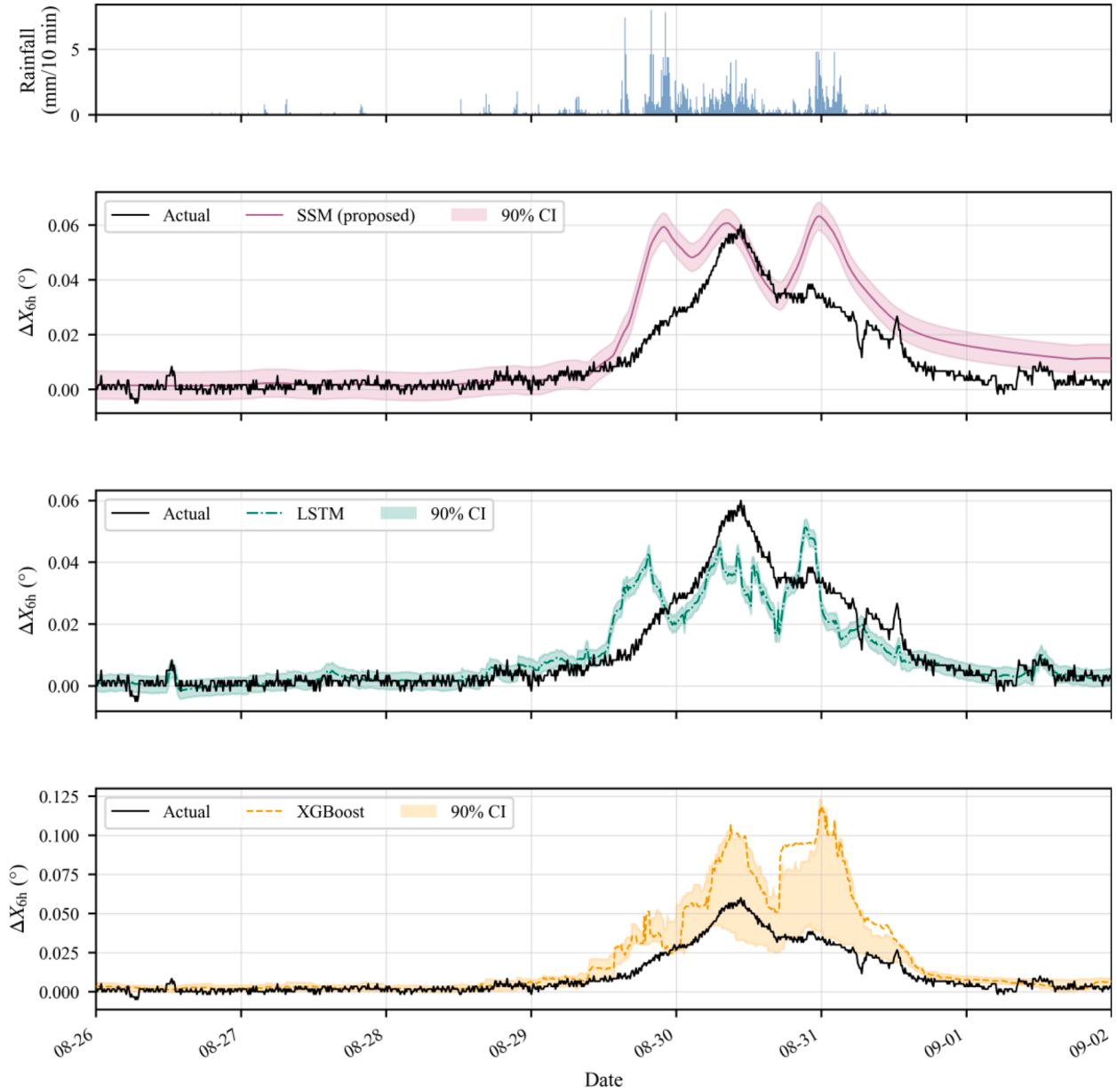


Fig. 12. Six-hour-ahead dX prediction comparison during sustained multi-pulse rainfall (2024-08-26 to 2024-09-02).

extremely narrow CI, indicating severe underestimation of uncertainty. XGBoost exhibits the largest magnitude bias, with predicted peaks reaching approximately 0.10–0.12 °, substantially exceeding the observed values; the CI is consistently biased high, and the observed tilt response falls below the lower CI bound for extended periods.

Taken together, both cases highlight the same mechanistic contrast. By explicitly tracking the slope's proximity to a critical state through the physical hidden states, SSM responds promptly to abrupt changes in rainfall intensity, maintains reasonable tilt response magnitude estimates under sustained rainfall, and delivers prediction intervals that are appropriately scaled to the forecast horizon through covariance propagation. The baseline methods exhibit systematic deficiencies in both event magnitude estimation and uncertainty quantification, and their failure modes point in opposite directions: LSTM tends to underestimate, whereas XGBoost tends to overestimate on this dataset. This divergence further indicates that the core challenge in this monitoring context is not curve fitting alone, but rather the capacity to represent the underlying physical process.

4.5. Hidden state visualisation

Fig. 13 presents a three-panel comparison of the SSM-learned hidden state h_{wet} (middle panel) against three manually constructed effective rainfall curves with half-lives of $M = 24, 48,$ and 72 h (bottom panel), with 1-hour aggregated rainfall shown in the top panel for reference. The observation period spans May to August 2024, and the model shown is SSM dX 6 h.

The physical interpretation of h_{wet} is that of a learnable effective rainfall variable. In the SSM state transition equation, h_{wet} decays exponentially at rate a_{wet} during rain-free periods, and its recurrence is fully equivalent to the classical effective rainfall formulation:

$$h_{\text{wet}}(t) \approx \sum_{\tau \leq t} R_{\tau} \cdot a_{\text{wet}}^{t-\tau} \quad (22)$$

Unlike XGBoost, which requires 72 half-life values to be prescribed manually, SSM learns a_{wet} directly from data. The equivalent half-life is recovered as:

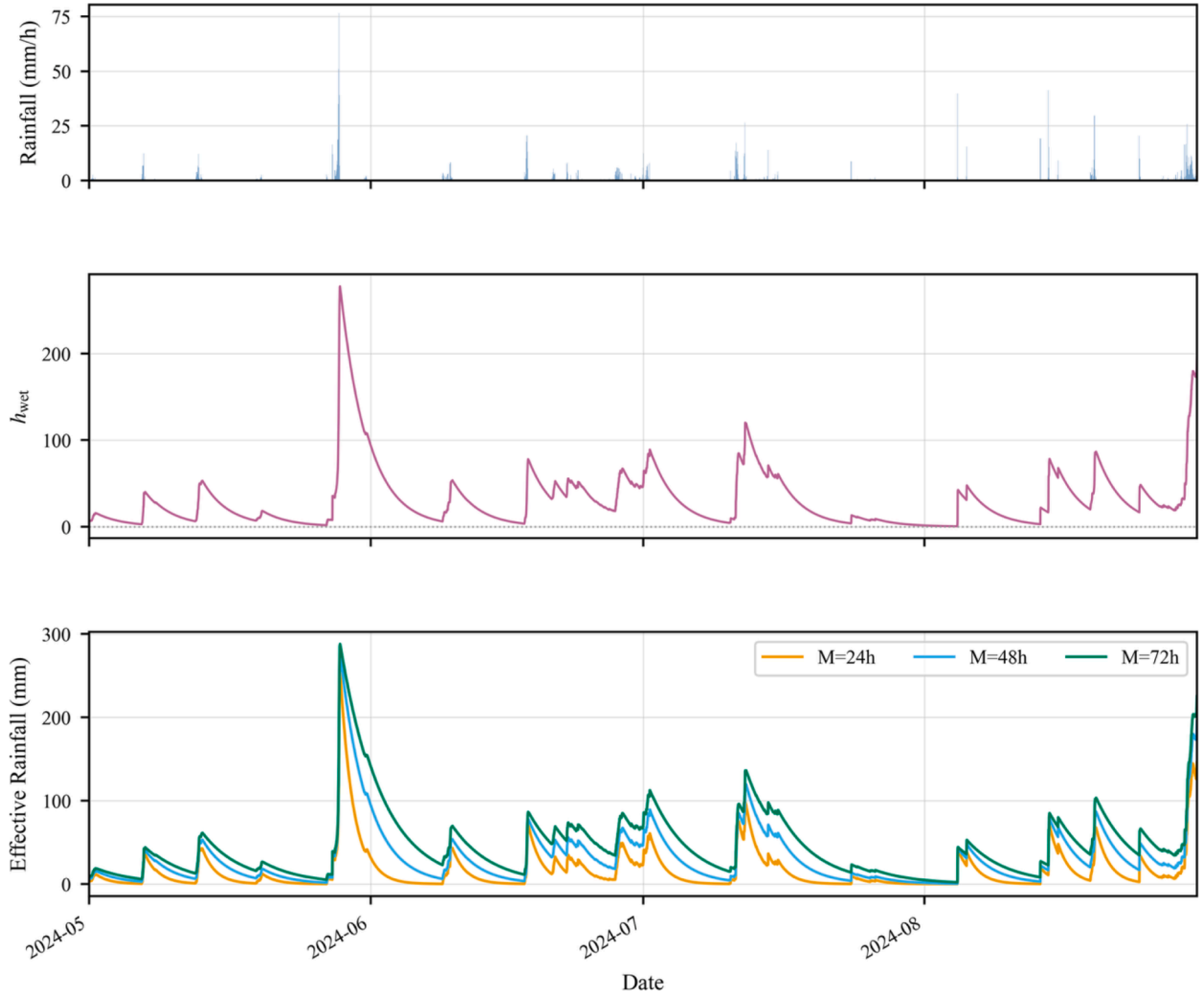


Fig. 13. Learned hidden state h_{wet} compared with manually constructed effective rainfall curves (half-lives $M = 24, 48, 72$ h).

$$M = \frac{\ln 0.5}{6 \times \ln a_{wet}} \quad (23)$$

For the study slope, the learned parameter corresponds to a half-life of 47.61 h, implying that the soil at this site retains a hydrological memory of approximately two days. Fig. 13 shows that the decay waveform of h_{wet} most closely matches the $M = 48$ h curve. This differs from XGBoost, where 72 half-life values are prescribed a priori as fixed input features (Eq. (19)); in the SSM, the half-life is recovered from the learned parameter a_{wet} using Eq. (22). The agreement with the empirical $M = 48$ h effective-rainfall formulation used in Japanese slope engineering suggests a physically plausible memory scale. However, because no independent pore-water pressure or soil-moisture measurements are available at this site, this value should be regarded as a data-driven estimate rather than a confirmed site property.

5. Discussion

5.1. Comparison between separate dx/dy prediction and direct resultant tilt response prediction

In this study, dX and dY were modeled independently instead of directly predicting the resultant tilt response $dR = \sqrt{dX^2 + dY^2}$, primarily because the latter is incompatible with the Gaussian observation-noise assumption underlying the EKF. The fundamental reason lies in noise-model compatibility. The SSM is derived on the basis of an

Extended Kalman Filter (EKF), whose observation noise is assumed to be zero-mean, additive, and Gaussian, with support over $(-\infty, +\infty)$. Because dX and dY take both positive and negative values, they naturally satisfy this assumption. By contrast, $dR \geq 0$ by definition, so the error distribution is truncated near $dR \approx 0$ and is no longer symmetric. This truncation causes the Gaussian assumption to break down precisely in the low-tilt response regime, introducing systematic bias into both the covariance propagation and the likelihood evaluation of the EKF (Thrun et al., 2005).

Beyond noise-model compatibility, separate modeling expands the prediction degrees of freedom from one (the scalar dR) to two (the vector (dX, dY)). At comparable prediction accuracy, this allows the tilt response azimuth to be recovered as $\hat{\theta}(t) = \arctan(\hat{dY} / \hat{dX})$, providing richer diagnostic information for engineering early-warning systems. Analysis of the test set shows that the distribution of predicted tilt response azimuths is concentrated around 22.3° (equal to the observed mean), indicating that the two independently trained models implicitly preserve directional consistency at the physical level.

5.2. Model limitations

Independent modeling per prediction horizon. Separate SSM parameter sets were trained for each of the 24 prediction horizons (1 through 24 h). This design reflects the fact that the mechanisms by which antecedent rainfall drives soil saturation accumulation differ

across time scales, and the optimal parameters (e.g., a_{wet} , c_{fut}) therefore vary accordingly. A single unified model would struggle to capture both short-term and long-term dynamics simultaneously. The trade-off is that the total number of parameters grows linearly with the number of horizons, and no explicit physical-consistency constraint is imposed across horizons.

The training and test periods both correspond to relatively stable creep displacement, and neither period contains a failure event. The model should therefore be interpreted as a tilt response prediction model: it forecasts sensor-level angular increments under rainfall forcing during stable creep. It is not a slope failure prediction model, which would need to identify the transition from stable creep to accelerating deformation and failure (Fukuzono, 1985; Saito, 1961). During large-scale slope failure, displacement rates can increase sharply and the rainfall–tilt–response relationship may become strongly nonlinear (Leroueil, 2001), so patterns learned from the present stable-creep data may no longer hold. Validation against historical failure-event records is required before the model can be used for near-failure assessment. Future work will extend the state-space formulation by incorporating creep mechanics for failure-stage deformation analysis.

Future rainfall is represented as a scalar mean, which limits the ceiling accuracy at long horizons. The variable r_{fut} is defined as the mean rainfall over the next H hours (a scalar), and consequently discards information about the temporal distribution and peak position of rainfall within that window. In operational deployment, errors in numerical weather prediction forecasts propagate directly through the coefficient c_{fut} into tilt response prediction (Rodríguez-Rincón et al., 2015); the larger the forecast error and the longer the horizon, the greater the degradation in predictive accuracy. The recommended primary application window for this model is 5 to 8 h ahead, within which forecast-mean bias remains relatively manageable and both prediction accuracy and uncertainty estimates remain operationally useful. A natural next step is to replace the scalar-mean input with a complete rainfall time-series sequence, but this extension would require a higher-capacity architecture and separate validation.

5.3. Sensitivity to future rainfall input error

In operational deployment, r_{fut} must come from weather forecasts and will contain errors. The test-set r_{fut} was multiplied by $\alpha_r \in [0.5, 1.5]$ at 0.1 intervals, and R^2 and 90% PI coverage were re-evaluated at $H = 3, 8, 12$, and 24 h for dX without retraining (Fig. 14).

Short horizons are robust. R^2 remains within 0.55–0.77 ($H = 3$ h) and 0.49–0.86 ($H = 8$ h) over the full perturbation range, with PI coverage above 90% throughout, because predictions at these horizons are dominated by the autoregressive state-propagation terms rather than

$c_{\text{fut}} \cdot r_{\text{fut}}$.

Long horizons degrade rapidly and asymmetrically. At $\alpha_r = 1.2$, R^2 drops from 0.86 to 0.14 ($H = 12$ h) and from 0.82 to 0.29 ($H = 24$ h). The equivalent underestimation ($\alpha_r = 0.8$) reduces R^2 to only 0.73 and 0.80. PI coverage at $H = 12$ h falls below 80% once α_r exceeds 1.2.

These results reinforce the recommended application window of 5–8 h (§5.2): within that range, even $\pm 50\%$ rainfall errors cause only modest accuracy loss. Beyond 8 h, prediction quality becomes contingent on the rainfall forecast. Coupling with ensemble rainfall forecasts is a direction for future work.

5.4. Cross-Site transferability

Validation in this study is based on a single site. Three structural elements carry over without modification: the two-state decomposition (h_{wet} driving h_{inst}), the power-softplus observation function, and the Delayed-EKF training procedure.

All 8 learnable parameters must be re-estimated from site-local data, as they encode drainage rate, coupling strength, dissipation, and noise levels governed by local geology, hydrology, and sensor geometry. Because the model has only 8 parameters, each with outsized influence, conventional transfer-learning strategies are inapplicable. Deployment requires tilt sensors and a rain gauge, records covering at least 10 distinct rainfall events, and L-BFGS-B optimization of the 8 parameters. Periodic re-estimation is recommended as observation data accumulate.

The prediction outputs support an operational early-warning workflow combining two information sources: real-time monitoring, where sustained elevation of observed tilt increments signals a transition from primary to secondary creep; and SSM forecasts, which predict the tilt increment and 90% PI over the next H hours, providing advance warning before a critical response materializes. The warning level is determined jointly by both sources. A missed alarm occurs when the slope enters accelerated creep but neither source triggers a warning; a false alarm is the converse. Quantitative evaluation of alarm rates requires labeled event records from extended deployment, which is a direction for future work.

6. Conclusions

This study developed a physics-based framework for multi-horizon prediction of rainfall-induced slope response by combining a two-state-space model (SSM) with an Extended Kalman Filter (EKF). Using tilt-monitoring data from a slope site in Japan, the proposed framework was systematically evaluated against a tree-based ensemble model (XGBoost) and a sequence model (LSTM). The main conclusions are as follows.

1. First, explicit modeling of physical latent states is central to improving predictive performance in this low-dimensional monitoring setting. With only 8 learnable parameters, the proposed SSM consistently outperformed both baselines across forecast horizons from 1 to 24 h for both dX and dY, achieving a peak R^2 of 0.857 for dX at the 12 h horizon. This result suggests that, under rainfall-driven monitoring conditions, predictive performance depends less on model size than on whether the underlying physical process is represented in a structurally meaningful way.
2. Second, analytical covariance propagation within the EKF provided more reliable uncertainty estimates than those of the compared baselines. The 90% prediction-interval coverage of the SSM remained stable at 91–93% for dX, while the interval width expanded adaptively with forecast horizon. By contrast, XGBoost showed clear under-coverage at long horizons, and the residual-based intervals of LSTM did not adequately reflect horizon-dependent uncertainty. These results indicate that uncertainty quantification is not merely an auxiliary output, but an essential component of slope early-warning models.

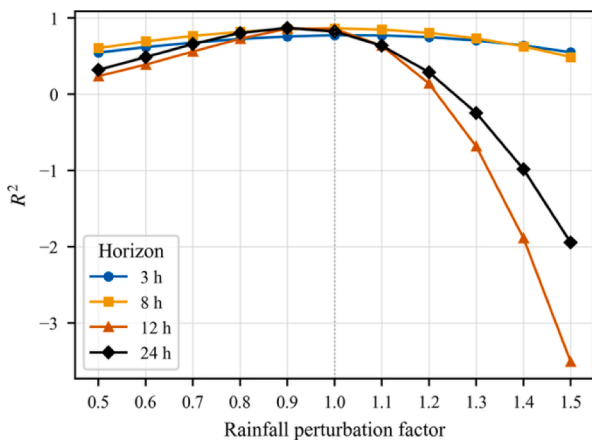


Fig. 14. R^2 versus rainfall perturbation factor α_r (0.5–1.5) for $H = 3, 8, 12$, and 24 h (dX).

3. Third, the learned model parameters were consistent with the hydrological characteristics of the monitoring site. In particular, the decay coefficient of the soil saturation state converged to an effective rainfall half-life of 47.61 h, which is consistent with the site's drainage behavior. This supports the physical interpretability of the proposed two-state structure and suggests that the framework may also provide useful information for site-specific hydrological characterization. Considering both predictive accuracy and uncertainty estimation, a forecast horizon of 5–8 h appears to be the most practical operating range for this site.

This study is limited to a single monitoring site, and cross-site generalizability remains to be examined. In addition, the available dataset does not include large-scale failure events, so the applicability of the model under strongly nonlinear failure conditions has not yet been validated. Future work should further examine model transferability and incorporate full rainfall time-series forecasts to improve long-horizon prediction performance.

CRedit authorship contribution statement

Yupeng Cao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Hideaki Yasuhara:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tomoki Nakazora:** Writing – review & editing, Methodology, Investigation, Data curation. **Weiren Lin:** Writing – review & editing, Writing – original draft, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was financially supported by Ministry of Land, Infrastructure, Transport and Tourism, Japan, Technical Research and Development for Road Policy Quality Improvement and Ehime Prefecture through the Digital Implementation Acceleration Project “TRY ANGLE EHIME”. The authors used AI language tools to assist with English editing during the preparation of this manuscript.

Data availability

The authors do not have permission to share data.

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